

SUMMARY REPORT

Project: Transportation Systems

Client: Federal Highway Administration (FHWA)

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Introduction

This initiative was started by the Federal Highway Administration (FHWA) in order to use high-fidelity vehicle trajectory data to better understand microscopic traffic flow features. This particular analysis's goal is to identify and display patterns of vehicle interaction, with a particular emphasis on car-following behavior. We can improve safety procedures for automated driving systems and better calibrate traffic simulation models by looking at how individual vehicles react to the actions of those directly in front of them.

Methods

Variables Selected and Rationale

The team focused on the following key variables from the **NGSIM.csv** dataset:

- **trajectory_number**: Unique identifier for each vehicle's trip through the study area. We selected Trajectory #1 to focus on a single, complete vehicle path for detailed analysis.
- **Time**: Timestamp in seconds, used as the x-axis for all time-series plots. This provides the temporal framework needed to observe dynamic changes.
- **leader_speed(m/s)**: Speed of the preceding (lead) vehicle in meters per second.
- **follower_speed(m/s)**: Speed of the subject (following) vehicle in meters per second.

These variables were chosen because car-following is a fundamental microscopic traffic behavior. Leader and follower speeds allow direct observation of how closely the follower matches or reacts to the leader, which is essential for calibrating car-following models (e.g., Gipps, Intelligent Driver Model) used in simulation software. Other variables (e.g., position, acceleration, headway) exist in NGSIM but were not needed for this initial speed-based visualization.

Plot Construction

All plots were constructed in Python using **pandas** for data loading/filtering and **matplotlib.pyplot** for visualization:

1. Data Loading:

- `nhts_df = pd.read_csv('NHTS.csv')` (loaded but unused in this analysis).
- `ngsim_df = pd.read_csv('NGSIM.csv')`.

2. Data Preparation:

- Filter for one vehicle: `single_car = ngsim_df[ngsim_df['trajectory_number'] == 1].sort_values(by='Time')`. This isolates Trajectory #1 and ensures chronological order.

3. Leader Speed Plot:

- A new figure is created with `plt.figure(figsize=(10, 5))`.
- Line plot: `plt.plot(single_car['Time'], single_car['leader_speed(m/s)'], color='blue', label='Lead Vehicle')`.
- Title, x-label ("Time (seconds)"), y-label ("Speed (m/s)"), dashed grid, and legend are added.
- `plt.show()` displays the plot.

4. Follower Speed Plot:

- Identical structure, but using `single_car['follower_speed(m/s)']` and green color for the line.

No advanced statistical methods (e.g., regression or filtering) were applied; the focus was on straightforward time-series visualization to reveal raw behavioral patterns.

Results and Discussion

Key Findings from the Plots

- **Leader Speed Plot:** The blue line shows fluctuations in the lead vehicle's speed over time. Periods of steady speed alternate with accelerations and decelerations, typical of real highway traffic influenced by surrounding vehicles, congestion, or driver decisions.
- **Follower Speed Plot:** The green line for the following vehicle generally tracks the leader's speed but often with a slight delay or smoother transitions. Sharp drops or rises in leader speed are usually followed (with some lag) by similar changes in follower speed.

Overall, the plots illustrate **reactive car-following**: the follower attempts to maintain a safe distance and match the leader's pace, but human drivers introduce variability and response lag.

Relevance to the Transportation Industry

These findings have direct applications for FHWA and the broader transportation sector:

- **Traffic Simulation Models:** Observed reaction delays and speed-matching behavior can be used to calibrate and validate microscopic models, improving the accuracy of tools like VISSIM or SUMO for predicting congestion, safety, and capacity.
- **Safety Analysis:** Understanding how closely followers track leaders helps identify risky situations (e.g., sudden braking leading to rear-end collisions) and supports development of collision avoidance systems in connected and automated vehicles (CAVs).
- **Intelligent Transportation Systems (ITS):** Insights into real driver behavior inform adaptive cruise control (ACC) algorithms and variable speed limits, potentially reducing stop-and-go waves and improving highway flow.
- **Policy and Design:** FHWA can use such analyses to refine guidelines for highway design, signage, and traffic management strategies that account for human factors in mixed traffic environments.

Future work could extend this by combining both speeds on one plot, calculating time/space headways, or analyzing multiple trajectories to quantify average reaction times.

References:

Ahmed Project 3.ipynb. (2026). Vehicle Trajectory Analysis Codebase. Data sourced from FHWA NGSIM and NHTS datasets.